

Ultimate Windows, Linux & DevOps Command Guide

Comprehensive reference for students, developers, system administrators, and cybersecurity learners.

Windows CMD Basics

dir — List files and folders
cd — Change directory
tree — Display directory structure
mkdir — Create folder
rmdir — Remove folder
copy — Copy files
move — Move files
del — Delete files
ren — Rename files
type — View file contents
cls — Clear screen

Windows Administration

systeminfo — System details
tasklist — Running processes
taskkill /PID — Kill process
shutdown /s /t 0 — Shutdown
shutdown /r /t 0 — Restart
sfc /scannow — Repair system files
chkdsk — Disk check
wmic — Management interface

Windows Networking

ipconfig — IP configuration
ipconfig /all — Detailed network info
ping — Connectivity test
tracert — Trace route
nslookup — DNS lookup
netstat -ano — Network connections
arp -a — ARP table
route print — Routing table

PowerShell Essentials

Get-Process — List processes

Get-Service — List services

Get-ChildItem — List files

Set-Location — Change directory

Get-Content — Read file

Copy-Item — Copy file

Remove-Item — Delete file

Get-ComputerInfo — System info

Linux Basics

pwd — Current directory

ls -la — Detailed listing

cd — Change directory

mkdir — Create directory

rmdir — Remove empty directory

rm -rf — Delete directory

cp — Copy files

mv — Move files

touch — Create file

cat — Read file

nano — Edit file

Linux File Permissions

chmod 755 file — Change permissions

chmod +x script.sh — Make executable

chown user:file — Change ownership

umask — Default permissions

Linux System Administration

uname -a — Kernel info

hostnamectl — Host info

top — Resource monitor

htop — Interactive monitor

ps aux — Processes

kill -9 PID — Terminate process

systemctl status — Service status

journalctl — System logs

Linux Networking

ip a — IP addresses
ifconfig — Network config
ping — Connectivity test
netstat -tulnp — Ports
ss -tulnp — Socket statistics
traceroute — Trace route
dig — DNS lookup
curl — Transfer data
wget — Download files

Git Commands

git init — Initialize repository
git clone URL — Clone repo
git status — Repository status
git add . — Stage files
git commit -m — Commit changes
git push — Upload changes
git pull — Download changes
git branch — List branches
git checkout — Switch branch
git merge — Merge branches

Docker Commands

docker ps — Running containers
docker images — List images
docker pull — Download image
docker run — Run container
docker stop — Stop container
docker rm — Remove container
docker exec -it — Open shell

MySQL Commands

SHOW DATABASES; — List databases
CREATE DATABASE — Create database
USE db; — Select database
SHOW TABLES; — List tables
SELECT * FROM table; — Read data
INSERT INTO — Insert records
UPDATE — Update records

DELETE FROM — Delete records

Cybersecurity & Networking Tools

nmap — Port scanning

whois — Domain lookup

ssh — Remote login

scp — Secure file transfer

tcpdump — Packet capture

wireshark — Packet analysis

AWS CLI Commands

aws configure — Configure CLI

aws s3 ls — List S3 buckets

aws ec2 describe-instances — List EC2

aws iam list-users — List IAM users

Command Line Tips

- Use Tab for auto-completion.
- Use history in Linux to view previous commands.
- Use Ctrl+C to stop a running process.
- Use man in Linux for documentation.
- Use command /? in Windows for help.